PassFrame
tuition

Fraction Operations

Year 4 Pt 2 — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: Work out 2/3 − 1/6.

Method & answer: Rewrite 2/3 as 4/6 (6 is a multiple of 3). Then 4/6 − 1/6 = 3/6 (= 1/2).

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. 1/3 + 1/6

2. Which fraction is equivalent to 1/4 with a denominator of 8? A) 1/8 B) 2/8 C) 4/8 D) 8/4

3. 3/4 − 1/8

4. Multiply 1/3 by 1/2.

5. What is 1/3 + 1/9? A) 2/9 B) 4/9 C) 1/12 D) 2/12

6. 5/8 − 1/4

7. A box has 1/3 red counters and 1/6 blue counters. What fraction of the counters are red or blue?

8. Select ALL the pairs below where one denominator is a multiple of the other: A) 1/3 and 1/9 B) 1/4 and 1/6 C) 1/5 and 1/10 D) 1/2 and 1/7

9. 1/2 + 1/6

10. Multiply 1/4 by 1/3. A) 1/7 B) 1/12 C) 2/12 D) 1/3

11. 7/8 − 1/4

12. 1/5 + 3/10

13. A ribbon is 1/2 m and another is 1/8 m. What is the total length?

14. What is 3/5 − 1/10? A) 2/10 B) 5/10 C) 6/10 D) 7/10

15. 1/4 + 5/8

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. 1/4 + 1/3

2. Work out 3/4 − 2/5. A) 1/20 B) 7/20 C) 1/9 D) 5/20

3. 5/9 − 1/3

4. Multiply 2/5 by 3/4.

5. What is 1/2 × 2/3, in simplest form? A) 2/6 B) 1/3 C) 2/5 D) 3/6

6. 7/10 − 1/2

7. A survey found 2/3 of pupils walk to school and 1/5 cycle. What fraction walk or cycle?

8. Select ALL the calculations below equal to 3/4 − 1/2: A) 1/4 B) 2/8 C) 3/6 D) 5/20

9. 2/3 + 1/4

10. Multiply 4/5 by 1/2, giving your answer in simplest form. A) 4/10 B)

11. 11/12 − 1/4

12. 3/8 + 1/3

13. A charity shop sells 3/5 of its stock in a sale and donates 1/4 of its stock to another shop. What fraction of stock is sold or donated?

14. Which of these equals 2/3 × 3/8, in simplest form? A) 6/24 B) 1/4 C) 1/3 D) 2/8

15. 5/6 + 1/4 − 1/3

2/5 C) 5/8 D) 4/7

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What is double 1/6, plus 1/3?

2. What is half of 2/3, minus 1/6?

3. Add 1/4 to 1/2, then halve the result. A) 3/8 B) 3/4 C) 1/8 D) 3/16

4. What is 5/6 take away double 1/6?

5. Select ALL of these that equal half of 2/3: A) 1/3 B) 2/6 C) 1/6 D) 2/3 × 1/2

6. A paint pot is 5/6 full. A decorator uses 1/3 of a pot for one wall, then 1/4 of a pot for a second wall. What fraction of the pot is left?

7. A charity target is reached in two stages: 3/4 of the target in stage one, then a further 1/5 in stage two. By what fraction has the charity missed its full target (1 whole)?

8. A recipe needs 2/3 cup of milk. A smaller recipe uses 3/4 of that amount. How much milk is needed for the smaller recipe? A) 1/2 cup B) 5/12 cup C) 6/12 cup D) 8/12 cup

9. A plot of land has 1/2 used for a house and 1/5 used for a garage. The rest is garden. What fraction of the plot is garden?

10. Multiply 3/4 by 1/3, then add 1/6 to your answer. What is the result?

11. A tank starts 1/4 full. A pump adds 1/2 tank of water, then 1/8 tank is drained out. Select ALL the statements that are true about the tank's final level: A) It is more than half full B) It is exactly 5/8 full C) It is exactly 1/2 full D) It is less than 3/4 full

12. A swimming pool is filled to 1/3 capacity. Two hoses each add 1/4 of the pool's capacity. What fraction of the pool's capacity is filled in total (assuming it does not overflow)?

13. A cyclist has ridden 3/5 of a route. She then rides a further 1/4 of the total route. What fraction of the route has she ridden in total? A) 4/9 B) 17/20 C) 4/20 D) 1

14. A chef has 11/12 kg of butter. He uses 1/3 kg for pastry and 1/4 kg for icing. How much butter is left?

15. What is 5/6 − 1/4, then multiplied by 2/3?